EVERY CLOSED SUBSCHEME OF A RELATIVE PROJ OVER A QUASI-COMPACT BASE IS ITSELF A RELATIVE PROJ OF A GRADED QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 7903-7917 and proof span 7919-7983 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains the author's phrase 'Having said this' and a reference to the proof of a cited constructions lemma for the description of the ideal J'; both are preserved.

Editorial note. 6 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The relative Proj of a quasi-coherent graded algebra is separated, and pushforward along a quasi-compact separated morphism preserves quasi-coherence, cited from the constructions and schemes chapters. Cited at proof lines 7919-7924. Makes the kernel of the canonical map of graded modules quasi-coherent, hence a legitimate graded quotient algebra.
- A surjection of quasi-coherent graded algebras induces an everywhere defined closed immersion of relative Proj's, cited from the constructions chapter. Cited at proof lines 7926-7929. Produces the closed subscheme Z' that has to be shown equal to Z.
- Quasi-compactness of a Proj over an affine base in terms of finitely many homogeneous elements generating the irrelevant ideal, cited from the constructions chapter. Cited at proof lines 7934-7937. Lets the proof reduce the comparison of two closed subschemes to the affine charts D_+(f_i).
- Invertibility of the twists of the structure sheaf on Proj and the identification of the multiplication maps of twists of a quasi-coherent module as isomorphisms. Cited at proof lines 7951-7956. Allows homogeneous elements to be adjusted by powers of the chart equation without changing the ideal they generate.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01MD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2118–2118. Same-chapter reference used in the retained author text at source line(s) 7684, 7738, 7939. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01O2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2179–2179. Same-chapter reference used in the retained author text at source line(s) 7644, 7921, 8227, 8466. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01N0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2141–2141. Same-chapter reference used in the retained author text at source line(s) 7960, 7990. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 07ZI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10148–10148. Same-chapter reference used in the retained author text at source line(s) 7931. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01MS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2133–2133. Same-chapter reference used in the retained author text at source line(s) 7956. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01LC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2081–2081. Same-chapter reference used in the retained author text at source line(s) 741, 6120, 7924. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{A}$ be a quasi-coherent graded $\mathcal{O}_S$-algebra. Let $p : X = \underline{Proj}_S(\mathcal{A}) \to S$ be the relative Proj of $\mathcal{A}$. Let $i : Z \to X$ be a closed subscheme. Denote $\mathcal{I} \subset \mathcal{A}$ the kernel of the canonical map*

$$\mathcal{A} \longrightarrow \bigoplus\nolimits_{d \geq 0} p_* \left((i_* \mathcal{O}_Z)(d) \right).$$

If p is quasi-compact, then there is an isomorphism $Z = \underline{Proj}_S(\mathcal{A}/\mathcal{I})$.

Proof. The morphism p is separated by Constructions, Lemma 01O2. As p is quasi-compact, p_* transforms quasi-coherent modules into quasi-coherent modules, see Schemes, Lemma 01LC. Hence $\mathcal{I}$ is a quasi-coherent $\mathcal{O}_S$-module. In particular, $\mathcal{B} = \mathcal{A}/\mathcal{I}$ is a quasi-coherent graded $\mathcal{O}_S$-algebra. The functoriality morphism $Z' = \underline{\mathrm{Proj}}_S(\mathcal{B}) \to \underline{\mathrm{Proj}}_S(\mathcal{A})$ is everywhere defined and a closed immersion, see Constructions, Lemma 07ZI. Hence it suffices to prove $Z = Z'$ as closed subschemes of X.

Having said this, the question is local on the base and we may assume that $S = \mathrm{Spec}(R)$ and that $X = \mathrm{Proj}(A)$ for some graded R-algebra A. Assume $\mathcal{I} = \widetilde{I}$ for $I \subset A$ a graded ideal. By Constructions, Lemma 01MD there exist $f_0, \ldots, f_n \in A_+$ such that $A_+ \subset \sqrt{(f_0, \ldots, f_n)}$ in other words $X = \bigcup D_+(f_i)$. Therefore, it suffices to check that $Z \cap D_+(f_i) = Z' \cap D_+(f_i)$ for each i. By renumbering we may assume $i = 0$. Say $Z \cap D_+(f_0)$, resp. $Z' \cap D_+(f_0)$ is cut out by the ideal J, resp. J' of $A_{(f_0)}$.

The inclusion $J' \subset J$. Let d be the least common multiple of $\deg(f_0), \ldots, \deg(f_n)$. Note that each of the twists $\mathcal{O}_X(nd)$ is invertible, trivialized by $f_i^{nd/\deg(f_i)}$ over $D_+(f_i)$, and that for any quasi-coherent module $\mathcal{F}$ on X the multiplication maps $\mathcal{O}_X(nd) \otimes_{\mathcal{O}_X} \mathcal{F}(m) \to \mathcal{F}(nd + m)$ are isomorphisms, see Constructions, Lemma 01MS. Observe that J' is the ideal generated by the elements g/f_0^e where $g \in I$ is homogeneous of degree $e \deg(f_0)$ (see proof of Constructions, Lemma 01N0). Of course, by replacing g by $f_0^l g$ for suitable l we may always assume that $d|e$. Then, since g vanishes as a section of $\mathcal{O}_X(e \deg(f_0))$ restricted to Z we see that g/f_0^d is an element of J. Thus $J' \subset J$.

Conversely, suppose that $g/f_0^e \in J$. Again we may assume $d|e$. Pick $i \in \{1, \ldots, n\}$. Then $Z \cap D_+(f_i)$ is cut out by some ideal $J_i \subset A_{(f_i)}$. Moreover,

$$J \cdot A_{(f_0 f_i)} = J_i \cdot A_{(f_0 f_i)}.$$

The right hand side is the localization of J_i with respect to $f_0^{\deg(f_i)}/f_i^{\deg(f_0)}$. It follows that

$$f_0^{e_i} g / f_i^{(e_i + e)\deg(f_0)/\deg(f_i)} \in J_i$$

for some $e_i \gg 0$ sufficiently divisible. This proves that $f_0^{\max(e_i)} g$ is an element of I, because its restriction to each affine open $D_+(f_i)$ vanishes on the closed subscheme $Z \cap D_+(f_i)$. Hence $g/f_0^e \in J'$ and we conclude $J \subset J'$ as desired. $\square$